2026 prelim waves answers

Answer pages for the 2026 prelim waves questions.
Whole pages are kept. A page that also holds another answer is included in full.
DHS, NYJC and VJC answers are the worked papers. EJC, HCI and RI answers are solution booklets.

2026 RI P2 Q2 p.2
2026 RI P3 Q4 p.3
2026 DHS P2 Q3 p.5
2026 EJC P2 Q5 p.10
2026 EJC P3 Q4 p.12
2026 EJC P3 Q6 p.14
2026 HCI P2 Q3 p.16
2026 HCI P2 Q4 p.17
2026 HCI P3 Q10 p.18
2026 NYJC P2 Q4 p.21
2026 VJC P3 Q3 p.24

2 **(a)** The waves have a <u>constant phase difference</u>.

(not accepted: same amplitude, wavelength, velocity)

(b)

$$x = \frac{\lambda D}{a}$$

$$= \frac{530 \times 10^{-9} (2.7)}{0.85 \times 10^{-3}}$$

$$= 1.6835 \times 10^{-3} = 1.7 \times 10^{-3} \text{ m}$$

(c) **(i)** Less light passes through the narrower slits resulting in <u>dimmer bright fringes</u>. There is no change in the fringe separation as slit width is unchanged.

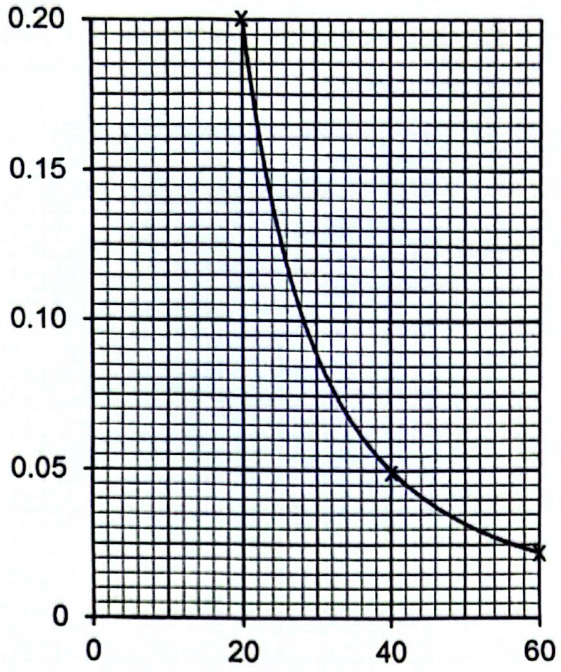
(note: 'decrease in contrast' is ambiguous. It must be clear that bright fringes are dimmer only. Dark fringes remain dark.)

(ii) As the screen is rotated, the distance between different parts of the screen to the double slit will not be the same.

The <u>fringe separations</u> will be <u>shorter</u> when the screen is <u>nearer</u> to the double slit, and <u>longer</u> on the part of the screen <u>further</u> away (fringe separation is proportional to slits-to-screen distance).

The fringes <u>when the screen is nearer to the double slits are brighter</u> and the fringes when the screen is <u>further away are dimmer</u> (light is spread over a larger area when the screen is further away).

$$I_{60} = \left(\frac{r_{20}}{r_{60}}\right)^2 I_{20}$$
$$= \left(\frac{20}{60}\right)^2 (0.20)$$
$$= 0.022 \text{ W m}^{-2}$$



All 3 points calculated and plotted correctly – 1M
Smooth curve – 1M
Deduct 1M for odd / inappropriate scale

Comments

4 (a) (i) Some students wrongly read off the graph and obtained the wrong value for $\cos^2\theta$ It is unnecessary to do so as the ratio of 0.50 is equal to $\cos^2\theta$.

(ii) Students should take care when calculating the new values of I_1 and draw the new line accurately. It is easier to use point(s) that fall(s) on the grid lines such as (0.86, 0.100) when drawing the new line. Do also take note to plot and read to half the smallest square.

(b) Some students did not calculate and plot accurately the values of I_d at 40 cm and 60 cm. In some cases, a small or odd scale was chosen for the intensity axis. Do label the intervals on the axis (every 10 small squares), without which the plotted points cannot be checked for accuracy.

5 (a) (i) $Q = CV = 120C\sin(314t)$

(ii) $I_C = \frac{dQ}{dt}$

$= \frac{d}{dt}(120\,C\sin(314\,t))$

$= 314(120\,C\cos(314\,t))$

$= 37700\,C\cos(314\,t)$

(b) (i) $I_{R,rms} = \frac{V_{rms}}{R} = \frac{120/\sqrt{2}}{200} = 0.424$ A

(ii) $P = \frac{1}{2}P_o \qquad (\text{or}: P = I_{rms}{}^2R)$

$= \frac{V_o^2}{2R}$

$= \frac{120^2}{2(200)}$

$= 36$ W

(c) (i) In a resistor, moving electrons continuously collide with the ionic lattice of the material. Changing the frequency changes how fast the current reverses direction, but it does not change the rate of these collisions. The mean rate of heat generated due to these collisions therefore remains the same, so r.m.s. current through the resistor remains constant.

Or:

r.m.s.current through the resistor remains the same, as $I_{R,rms} = \frac{V_o}{\sqrt{2}\,R}$ is constant/ independent of *f*.

(ii) When frequency increases, the capacitor charges and discharges much faster. Because the plates must fill up and empty out in a shorter period, the rate of flow of charge must increase. This increases the r.m.s. current through the capacitor.

Accept: r.m.s. current in the capacitor will increase, since $I_{C,rms} \propto \omega \propto f$.

Comments

5 (a) (i) Well done, a few erroneously expressed *C* instead of *Q* as subject of formula.

(ii) Well done, most correctly related I to rate of flow of charge *Q* and showed their workings fully to include the intermediate value 37680*C* cos(314*t*), which is commendable.

(b) (i) Generally well done, a few made the mistake of using 37700 or 37700*C* as peak current and obtained ridiculously huge $I_{R,rms}$ value. They did not recognise this is not the current for the resistor.

3 An experiment is carried out, using a light intensity meter attached to a toy train, to measure the diffraction of laser light through a double slit, as shown in Fig. 3.1 below.

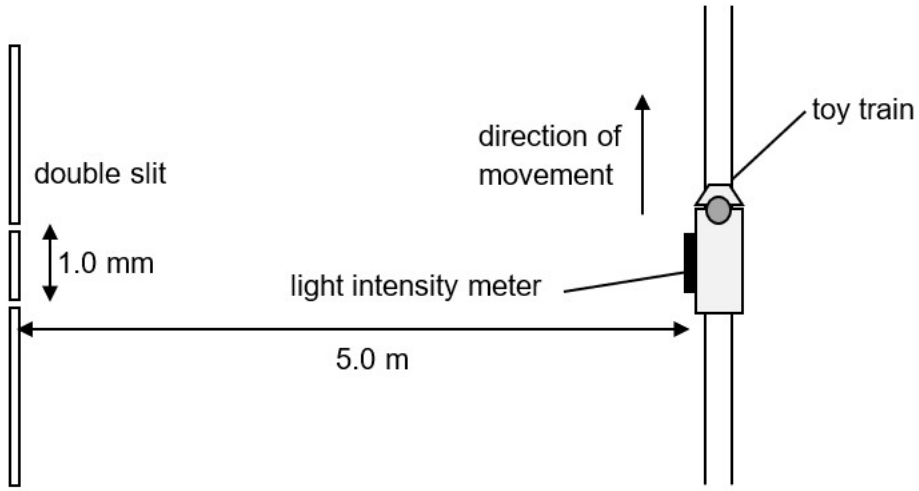


Fig. 3.1

Light from the laser is incident on the double slit. The light from the double slit is then detected by a light sensor attached to the train.

As the train moves along the track, the light intensity is recorded by the light intensity meter. The variation with time of the light intensity is shown in Fig. 3.2.

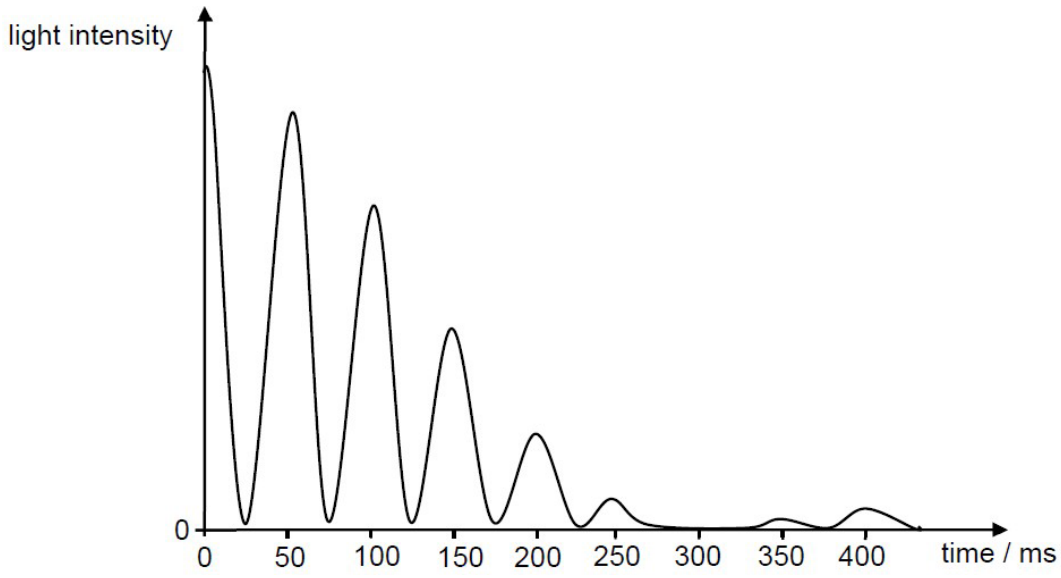


Fig. 3.2

(a) Explain, with reference to the paths travelled by the light emerging from each slit, why maxima and minima are observed.

Light from each slit travels a different distance to the sensor. When the path difference from each slit to the light intensity sensor is $n\lambda$ where n is an integer, the waves meet in phase and constructive interference occurs, light intensity reaches maximum value. [B1]

When the path difference from each slit to the light intensity sensor is $\left(n+\frac{1}{2}\right)\lambda$ where n is an integer, the waves meet antiphase and destructive interference occurs, light intensity reaches minimum value. [B1]

..... [2]

(b) **(i)** The speed of the train is 6.5 cm s^{-1}.
Use Fig. 3.2 to show that the fringe separation of the two slit interference pattern is 3.3×10^{-3} m.

fringe separation, $x = \left(\frac{200\times10^{-3}\ \text{s}}{4}\right)\times\left(6.5\times10^{-2}\ \text{m s}^{-1}\right) = 3.25\times10^{-3}\ \text{m}$ [B1]

(do not accept answer based on 50 ms for one fringe spacing directly)

[Students should measure over several fringe spacings and divide by the number of intervals to reduce the percentage uncertainty in the fringe separation.]

[1]

(ii) The separation of the slits is 1.0 mm and the track is 5.0 m from the slits.
Determine the wavelength of the laser light.

Since the slit-to-screen distance is much greater than the slit separation, i.e., $D \gg a$, we have $x = \frac{\lambda D}{a}$. Therefore,

$$\lambda = \frac{ax}{D} = \frac{\left(1.0\times10^{-3}\right)\left(3.25\times10^{-3}\right)}{5.0}$$
$$= 6.5\times10^{-7}\ \text{m}$$

wavelength = ... m [2]

(c) A maximum is detected by the light intensity meter at every 50 ms, explain why a minimum is detected at 300 ms in Fig. 3.2.



The double slit interference pattern is modulated / modified / enveloped by the single slit diffraction pattern.

OR There is an envelope because of (single) slit diffraction

At 300 ms the light sensor is at the first minimum of the single slit diffraction envelope where the light intensity is zero (due to destructive interference between wavelets from different parts of each slit), so no maximum interference fringe is observed at that position. [B1]

..... [1]

(d) Determine the width of one of the slits.

position of first minimum, $y = (300 \times 10^{-3}\ \text{s})(6.5 \times 10^{-2}\ \text{m s}^{-1}) = 0.0195\ \text{m}$

single-slit diffraction first minimum: $b \sin\theta = \lambda$

small angle approximation: $\sin\theta \approx \tan\theta = \frac{y}{D}$

$\therefore b\left(\frac{y}{D}\right) \approx \lambda$

$\Rightarrow b \approx \frac{\lambda D}{y} = \frac{(650 \times 10^{-9})(5.0)}{0.0195} = 1.7 \times 10^{-4}\ \text{m}$

width = ... m [2]

(e) The experiment is changed so that the toy car is now used to investigate the formation of a stationary sound wave as shown in Fig. 3.3 below.

The light sensor of the train is replaced with a sound sensor attached to a data logger. A loudspeaker emitting sound waves of constant amplitude and frequency is placed such that it faces a reflecting barrier. The train track is adjusted so that the train is moving away from the loudspeaker towards the barrier.

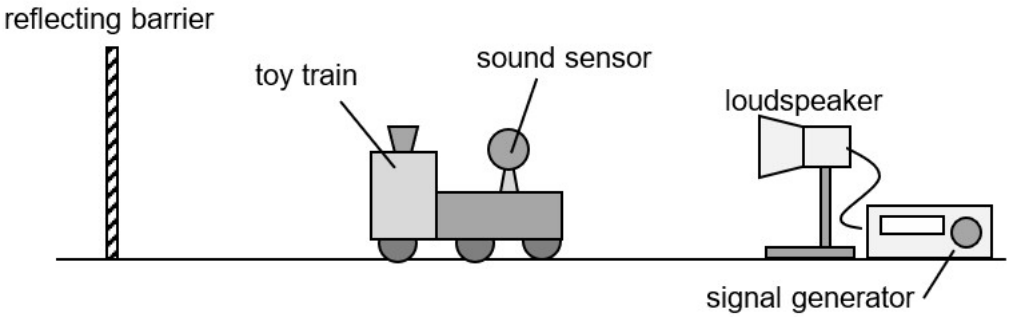


Fig. 3.3

The variation with distance of the sound intensity recorded by the sound sensor is shown in Fig. 3.4.

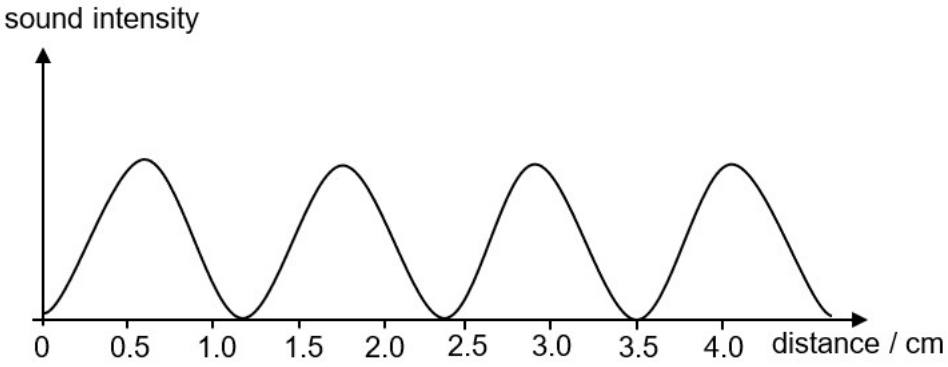


Fig. 3.4

(i) Explain how a stationary sound wave forms between the loudspeaker and reflecting barrier.

Sound wave from the loudspeaker is reflected by the barrier. [B1]

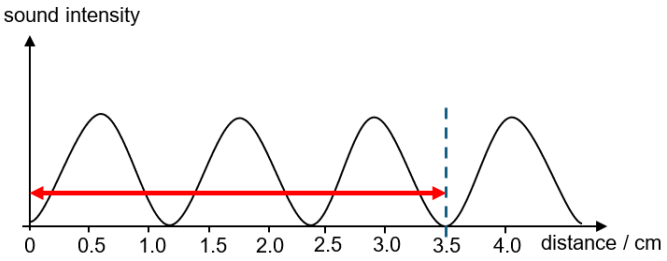
The reflected wave interferes / undergoes superposition with the incident wave. The resultant wave is a stationary wave. [B1]

…… [2]

(ii) Using Fig. 3.4, determine the wavelength of the stationary wave.

using separation between 4 consecutive nodes (1.5λ) as the distance from the graph, [B1]

wavelength, $\lambda = (3.5 \times 10^{-2}\ \text{m}) \div 1.5 = 0.0233\ \text{m}$ [A1]



wavelength = ... m [2]

(iii) Given that the frequency of the sound is 14 kHz.
Determine the speed of the sound wave.

speed, $v = f\lambda = (14000\ \text{Hz})(0.0233\ \text{m})$ [B1]

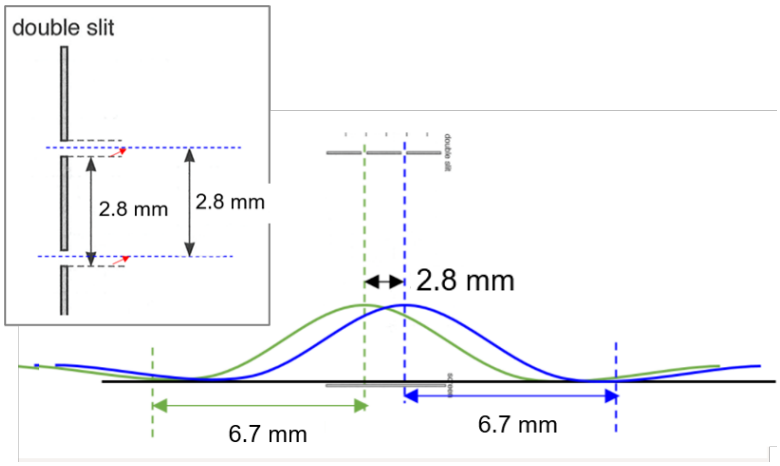
$= 326\ \text{m s}^{-1}$

$= 330\ \text{m s}^{-1}$ (to 2 sig. fig.) [A1]

speed = ... m s^{-1} [2]

[Total: 14]

Qns	Answer	Marks
4(b)	e.m.f. is the energy converted from non-electrical to electrical form per unit charge, while p.d. is the energy converted from electrical to non-electrical form per unit charge. [B1]	**B1**
4(c)	As R_2 increases, the total resistance of the circuit increases and the current in the cell decreases. The p.d. across the internal resistance therefore decreases. Hence, using $V_{terminal} = E - Ir$, the terminal p.d. increases. Since R_1 is constant, the current through R_1 increases. -	**M1** **M1** **A1**
4(d)(i)	p.d. across r = 1.40 − (0.220 × 5.50) = 0.19 V. Icell = 0.19/0.13 = 1.46 A ≈ 1.5 A (2 s.f.).	**C1** **A1**
4(d)(ii)	I_2 = 1.5 − 0.220 = 1.28 A; R_2 = (0.220×5.50)/1.28 = 0.945 Ω	**C1** **A1**
	TOTAL FOR QUESTION 4	**9**
5(a)	For a single-slit diffraction pattern, $\sin\theta = \frac{\lambda}{b} \Rightarrow \theta = \sin^{-1}\left(\frac{\lambda}{b}\right) = \sin^{-1}\left(\frac{670\times10^{-9}}{0.12\times10^{-3}}\right) = 0.00558336 \text{ rad (6.s.f)}$ Let the distance between the central maxima to first minima for the diffraction pattern be *y*. Hence, $\tan\theta = \frac{y}{D} \Rightarrow y = D\ (\tan\theta)$ $= 1.20\ (\tan(0.00558336))$ $= 0.00670010 \text{ m (6.s.f)}$ Thus, the width of the central fringe is: 2*y* = 2 (0.00670010) = 0.0134 m (3.s.f) = 13.4 mm (shown) **Note:** Deduct 1 mark if students - used small angle approximation without justification - write the angle in degrees without indicating the degree symbol	**C1** **C1** **M1** **A0**
5(b)(i)	The distance from the central bright maximum to first minimum is: 13.4 / 2 = 6.7 mm. However, the central bright maxima are 2.8 mm away from each other. Hence, we cannot distinguish between the two diffraction patterns from each single slit / they cannot be resolved.	**M1** **A1**

Qns	Answer	Marks
		
5(b)(ii)	$x = \frac{D\lambda}{a} = \frac{(1.2)(670\times10^{-9})}{(2.8\times10^{-3})} = 0.000287143 \text{ m (6.s.f)}$ An estimation of the number of fringes, *n* is $n = \frac{0.0134002}{0.000287143} = 46.66733 = 46$ (round down)	**C1** **A1**
	TOTAL FOR QUESTION 5	**7**
6(a)	Faraday's law of electromagnetism states that the (induced) e.m.f. is directly proportional to the rate of change of magnetic flux linkage.	**B1**
6(b)(i)	$\varepsilon = \frac{d\phi}{dt} = \frac{\Delta\phi}{\Delta t}$ $\Delta\phi = \varepsilon \times \Delta t$ $= 0.55 \times 14 = 7.7$ Wb Accepted units include: Wb / Vs / Tm^2	**A1** **A1**
6(b)(ii)	$\phi = BA$ $A = \frac{\phi}{B} = \frac{7.7}{38\times10^{-6}}$ $= 2.0\times10^{5}$ m^2 **Note:** Allow ECF from **(b)(i)**	**A1**
6(b)(iii)	**Method 1:** Area = speed x time x width $v = \frac{A}{t \times \text{width}}$ $= \frac{2.0\times10^{5}}{14\times68}$ $= 210$ m s^{-1}	**C1** **A1**

Qns	Answer	Marks
	Note: accept 73000 J and 73500 J.	
3(b)	$Q_{\text{ice at -8.0°C to ice at 0°C}} = m_{\text{ice}} c_{\text{ice}} \Delta T$ $= 500 \times 2.09 \times \left[0 - (-8.0)\right]$ $= 8360$ J (3 s.f.) **Note: accept 8360 J and 8400 J.**	**B1**
3(c)	0 °C Since the heat needed to raise the temperature of the original ice to 0 °C (= 8360 J) is greater than the heat released by water as its temperature decreases to 0 °C (= 4598 J), but is less than the total heat released by water as it freezes to 0 °C ice (= 4598 + 73480 = 78078 J), the original water will not fully freeze into 0 °C ice. The final state will be water-ice mixture at 0 °C. **Note:** When the temperature of the water decreases to 0 °C (releasing 4598 J of heat), the temperature of the original ice is still below 0 °C (to raise it to 0 °C, 8360 J of heat is needed). Hence, the original water will continue to partially freeze, until the original ice is at 0 °C. The final product is a water-ice mixture at 0 °C.	**A1** **M1**

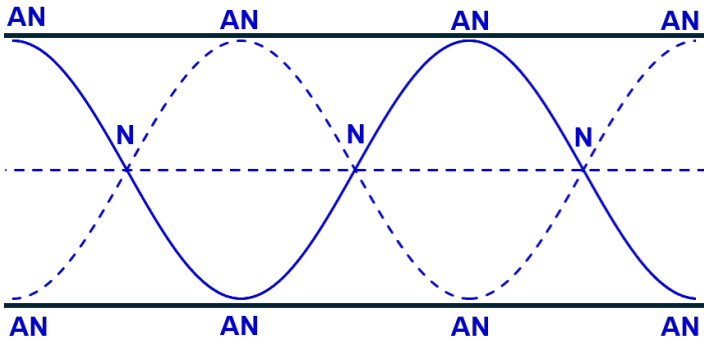
Qns	Answer	Marks
4(a)(i)	Power received = $\frac{\text{Area of source}}{4\pi r^2} \times$ Power of source $= \frac{450 \times 10^{-4}}{4\pi (3500)^2} \times 15.0$ [M1] $= 4.4 \times 10^{-9}$ W [A1]	
4(a)(ii)	Energy from the source radiates uniformly in all directions	**B1**

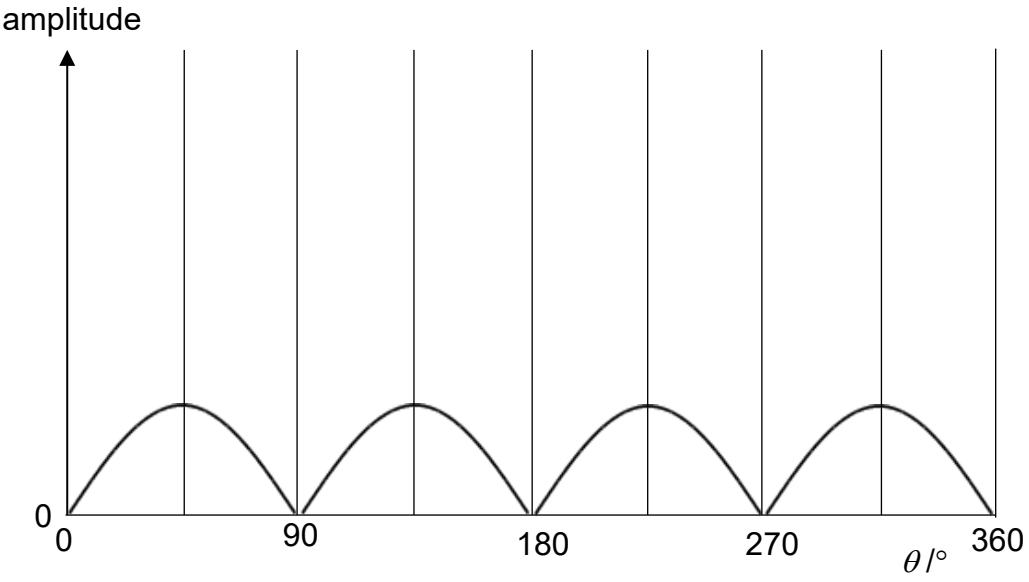
Qns	Answer	Marks
4(a)(iii)	Intensity $\propto$ Amplitude^2 Intensity = $\frac{\text{Power}}{4\pi r^2} \propto \frac{1}{r^2}$ Hence $A \propto \frac{1}{r}$ $\frac{A_{3.5}}{A_{2.5}} = \frac{r_{2.5}}{r_{3.5}}$ $\frac{A_{3.5}}{A_o} = \frac{2.5}{3.5}$ [M1] $A_{3.5} = 0.71 A_o$ [A1]	
4(b)(i)	Oscillations/ vibrations occur in a single direction (in a plane) that is perpendicular to the direction of transfer of energy/propagation.	**B1**
4(b)(ii)	Apply Malus' Law, $I_1 = I_0(\cos 40°)^2$ $I_2 = I_0(\cos 40°)^2(\cos 80°)^2$ **M1** $= 0.018 Io$ (2 s.f.) **A1**	

Qns	Answer	Marks
5(a)	When an object freefalls, the **ONLY** force acting on the object is its weight (= *mg*). **M1** Hence, the freefall acceleration $a = \frac{F}{m} = \frac{mg}{m} = g$ is equal to the gravitational field strength *g*. **A1** OR By definition of gravitational field strength, $g = F_{\text{graviational}}/m$ and from Newton's 2nd Law, for a freely falling body, the **ONLY** force acting is gravitational force hence $F_{\text{net}} = mg = ma$. Therefore, $a = g$.	
5(b)	$g = \frac{GM}{r^2}$ $M = \frac{gr^2}{G} = \frac{9.81 \times (6.37 \times 10^6)^2}{6.67 \times 10^{-11}} = 5.9679 \times 10^{24} = 5.97 \times 10^{24}$	**B1**

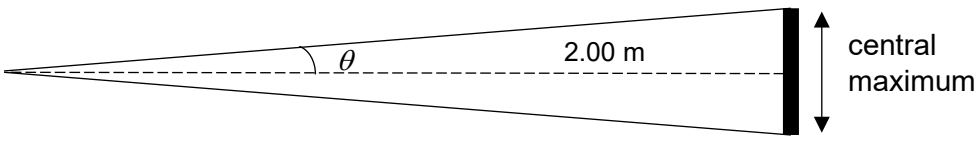
Qns	Answer	Marks
5(c)	$\phi = -\frac{GM}{r} = -\frac{6.67\times10^{-11}\times5.97\times10^{24}}{6.37\times10^{6}} = -6.25\times10^{7}\ \text{J kg}^{-1}$	B1
5(d)	The gravitational potential is taken to be zero at infinity. Since the gravitational force is attractive, the work done by the external force to bring the test mass from infinity to a point is negative. Hence, the gravitational potential is negative.	B1 B1 A0

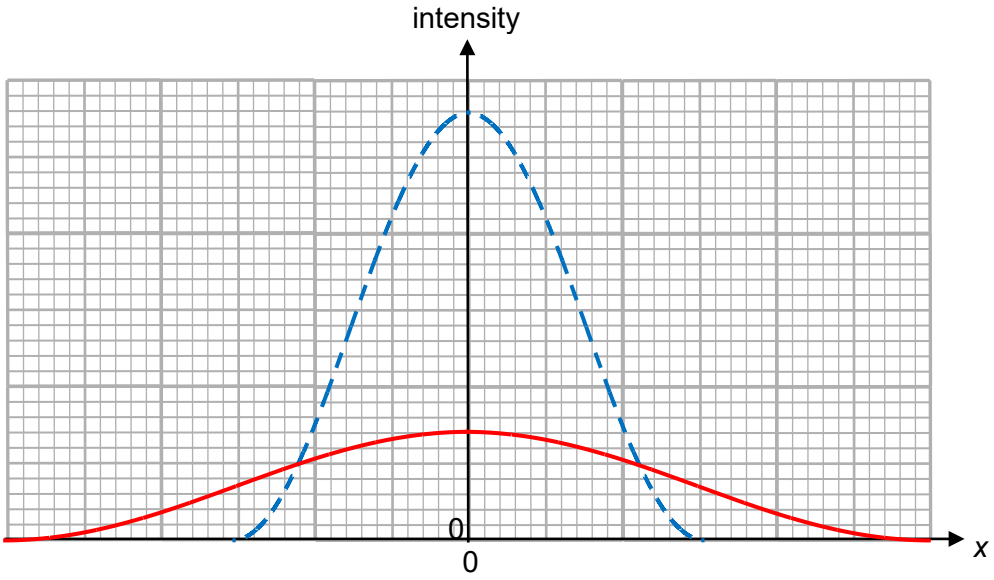
Qns	Answer	Marks
6(a)	The principle of superposition states that when two or more waves meet at a point, the resultant displacement is the sum of the individual displacements of each wave at that point. **Note:** Do **NOT** accept if students mention the "amplitude add up". The correct term is displacement.	B1
6(b)	Incident and reflected waves are of the same type, frequency and wavelength, and have the same speed, approximately the same amplitude and travel in opposite direction along the same tube. Incident wave from the loudspeaker reflects at the end / right side of the glass tube such that they superpose to form a stationary wave. **Note:** Do not accept if students write meet / overlap without superpose. The word interfere is not accepted as well.	B1 B1 B1
6(c)(i)	Amplitude of stationary wave: 0.06 + 0.06 = 0.12 mm	B1

Qns	Answer	Marks
6(c)(ii)	 General shape (include spacing) of a stationary wave (including dotted cosine curve – no solid line) with displacement antinodes at the ends of the glass tube. Stationary waves fulfilling the 3rd harmonic, n = 3, since there are three pressure antinodes (three displacement nodes) *#Although labelling **N** and **AN** is not a marking point, it is good practice to write it down.* **Note:** Students should know that a pressure antinode corresponds to a displacement node. Hence, there should be 3 displacement nodes which correspond to the 3rd harmonic for a pipe with two open ends.	**B1** **B1**
6(c)(iii)	$L = 1.5\lambda \Rightarrow \lambda = \frac{L}{1.5}$ $v = f\lambda \Rightarrow f = \frac{v}{\lambda} = \frac{v}{\left(\frac{L}{1.5}\right)} = \frac{320}{\left(\frac{15.0}{1.5}\right)} = 32 \text{ Hz}$	**M0** **A1**
7(a)(i)	${}^{210}_{83}Bi \rightarrow {}^{0}_{-1}e + {}^{210}_{84}Po$ + (anti) neutrino Accept beta particle	**B2**
(a)(ii)	total momentum/ total mass-energy/ total nucleon no/ total charge. Any two.	**A2**
(b)(i)	Mass of protons + neutrons = [(83 x 1.00729) + (127 x 1.00867)] u = 211.706 u Mass defect = 211.706 u - 209.939 u Mass defect = 1.767 u	**M1** **M1** **A0**

Question 3		
(a)	I_1 = 0.5 I_0 I_3 = 0.5 I_0 $\cos^2(25°)$ $\cos^2(65°)$ = 0.073 I_0 Note: award one mark if the student forgot to halve I_0 after passing through P_1 and gotten the final answer as 0.15 I_0.	**M1** **A1**
(b)	 **B1** – correct max and min positions **B1** – correct shape	

Question 4		
(a)	The minimas are nodes of the stationary wave / points where the incident and reflected waves meet in anti-phase / where destructive interference happen. The intensity of a wave emitted by a point source varies inversely with distance^2 / the amplitude of the wave varies inversely with distance. As plate moves rightward, the difference in intensity / amplitude between the incident and reflected waves becomes larger.	**B1** **B1** **B1**
(b)	Distance between nodes = 0.5 wavelength = 70 mm	**A1**

Question 10		
(a)	Coherent light waves are electromagnetic waves that have a **constant phase difference** between them.	**B1**
(b)(i)	Bright fringes are formed by the **constructive interference** of the coherent waves emerging from the two slits arriving on the screen **in phase**. [or with a constant phase difference of m x 2π rad where m: order (m= 0,1, 2 3…).] Dark fringes are formed by the **destructive interference** of the coherent waves arriving on the screen **out of phase.** [or with a constant phase difference of $(m-\frac{1}{2})\pi$ rad.] Note to marker: will only accept phase difference rather than path difference as phase difference is the fundamental condition for the outcome of the interference	**B1** **B1**
(b)(ii)	Using the fringe separation for double slit interference: fringe separation, $y = \frac{\lambda L}{d}$ $= \frac{(633 \times 10^{-9})(2.00)}{0.250 \times 10^{-3}}$ $= 5.064 \times 10^{-3}$ m Distance between the central bright fringe and the 2nd order dark fringe = 1.5y = 1.5 x 5.064 x 10^{-3} = 7.596 x 10^{-3} m = 7.60 mm	**C1** **A1**
(b)(iii)	The double-slit interference fringes are modulated by the single-slit diffraction envelope resulted from the appreciable width of each slit. As the single-slit diffraction envelope has its greatest intensity at the centre, the maximum observed at O is of a greater intensity than any other observable maximum.	**B1** **B1**
(c)(i)	The angle θ of first minimum for the single slit is given by: $\sin\theta = \frac{\lambda}{b}$ $\sin\theta = \frac{633\times10^{-9}}{0.080\times10^{-3}} = 0.0079125$ For small angle θ, $\sin\theta \approx \tan\theta$  $\therefore$ width of the central maximum = 2 x 2.00 (tan θ) = 2 x 2.00 x 0.0079125 = 0.03165 m = 31.7 mm	**B1** **B1** **A0**

	Note to marker: Students who do not use small angle approximation will have the same value.	
(c)(ii)	 Marking point: • Peak intensity is smaller. (1/4 of original intensity not required) • Width doubles	**A1** **A1**
(c)(iii)	Using the Rayleigh criterion for the resolving power of a single aperture $\theta_m \approx \frac{\lambda}{b}$ Using small angle approximation for the angle subtended by the two sources at the aperture, θ_m (in rad) $\approx \frac{s}{D}$ $\Rightarrow \frac{s}{D} = \frac{\lambda}{b}$ $\therefore D = \frac{s \times b}{\lambda} = \frac{(5.0 \times 10^{-3})(0.080 \times 10^{-3})}{633 \times 10^{-9}}$ $= 0.632$ $= 0.63$ m	**B1** **C1** **A1**
(d)(i)	Maximum angle of diffraction $\theta = \tan^{-1}\left(\frac{3.00}{2.00}\right) = 56.31^o$ From the grating equation $d \sin\theta = n\lambda$, $n = \frac{\sin\theta}{N\lambda}$ where N is the number of slits per metre $n = \frac{\sin 56.31^o}{(500 \times 10^3)(633 \times 10^{-9})} = 2.6$ Hence maximum observable order = 2 Maximum number of observable bright fringes = 2(2) + 1 = 5	**B1** **B1** **C1** **A1**

(d)(ii)	The **first order maximum** shows a **continuous spectrum of colours** (with red furthest from the central maximum and violet closest to the central maximum) while **the central maximum shows a** (narrow) **white line/spot.** From the diffraction equation $d \sin\theta = n\lambda$. For $n = 0$, all the wavelengths overlap, producing the white line/spot in the central maximum. In the first order maximum ($n = 1$), **different wavelengths deviate with different angles** θ with the longer wavelength deviating with the larger angle hence producing the spread of colours in the continuous spectrum.	**B1** **B1**

4 (a) (i) State **one** difference and **one** similarity between the oscillations of a stationary sound wave and a progressive sound wave.

Difference: ● In stationary sound waves, particles oscillate (except at the nodes where it is 0) with varying amplitudes from 0 to a maximum at the antinodes. In progressive sound waves, all particles oscillate with the same amplitude.

● In stationary sound waves, particles between adjacent nodes oscillate in phase while particles in adjacent segments oscillate in antiphase. In progressive sound waves, particles oscillate at different phases within one wavelength [B1] Differences must be in terms of phase or amplitude variations of a particle's oscillation NOT in terms of the whole wave (e.g. comparing energy transfers)

Similarity: In both waves, particles are oscillating in simple harmonic motion/about their equilibrium positions with the same frequency as the wave [B1] ignore longitudinal or cannot be polarised

(ii) Fig. 4.1 represents the displacement-time graph of two progressive sound waves travelling together through the same space.

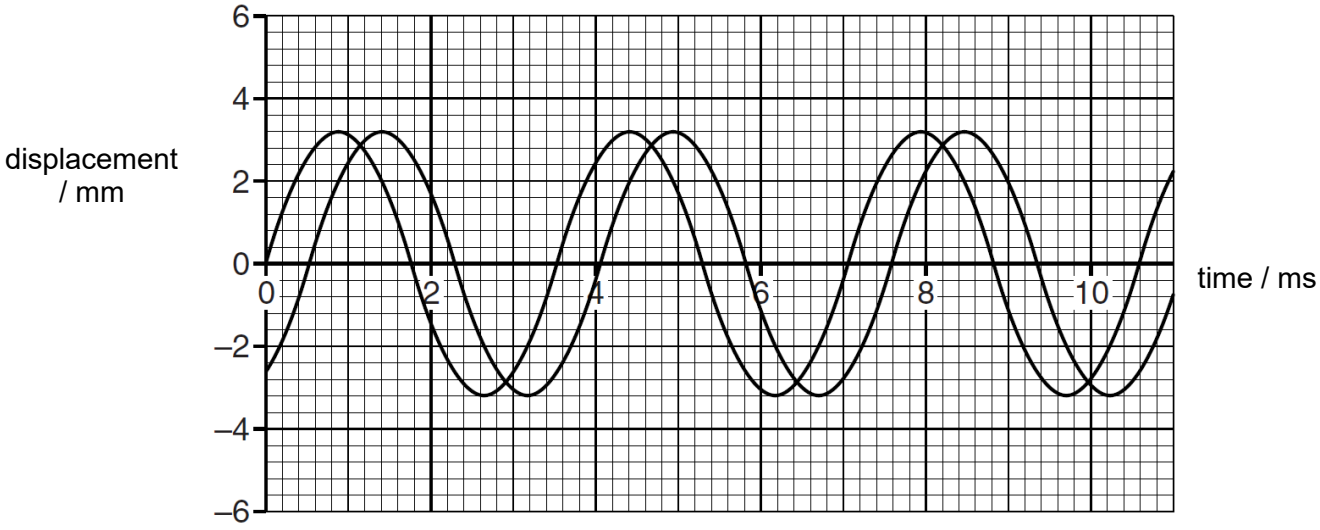


Fig. 4.1

1. State and explain whether Fig. 4.1 shows that the waves are coherent.

The waves are coherent because there is a constant time lag / constant period / constant frequency which indicates a constant phase difference [B1] apply evidence from graph of Fig. 4.1 to show coherence and not just mention "constant phase difference" only. [1]

NOT "same distance between the crests" since axis is given in time

2. Use Fig. 4.1 to determine the phase difference between the waves.

$$\Delta\phi = \frac{\Delta t}{T} \times 2\pi$$

$$= \frac{0.5}{3.5} \times 2\pi \quad \text{OR} \quad \frac{2.5 \text{ sq}}{17.5 \text{ sq}} \times 2\pi$$

$$= 0.898 \text{ rad} \quad [A1]$$

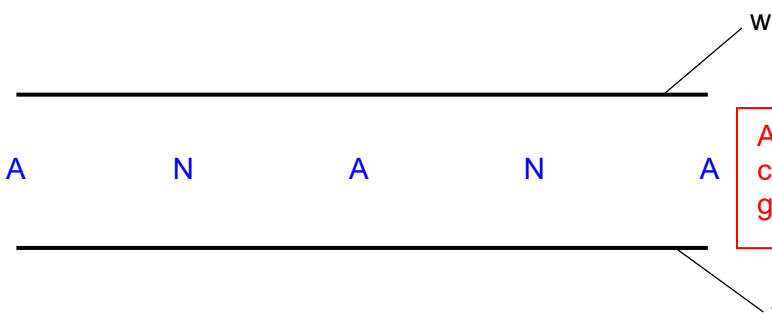
S.f. penalty for answers in fractions or in terms of π

phase difference = ... rad [1]

(iii) A sound wave travels along the axis of a tube that is open at both ends, as shown in Fig. 4.2. A stationary sound wave is formed in the tube.

The lowest frequency possible for the stationary sound wave in this tube is *f*.

On Fig. 4.2, mark all the positions of the displacement nodes (use the letter N) and the displacement antinodes (use the letter A) of a stationary sound wave for a frequency of 2*f*.



ALL N and A must be clearly marked (use the given symbols and not 'AN')

Fig. 4.2

[1]

(iv) With reference to the movement of air molecules on either side of a displacement node, explain why there is maximum variation in pressure at the nodes.

Particles on either side of a displacement node are moving in opposite directions [B1] ignore antinodes so at one moment the pressure increases to a maximum (particles on either side are closest to node) AND at the next moment the pressure decreases to a minimum (particles on either side furthest from node) therefore pressure varies from a maximum to minimum → hence maximum variation [B1] idea of pressure to the separation of neighbouring particles of the node. Ignore compressions and rarefactions only. [2]

(b) Fig. 4.3 shows the arrangement for an experiment involving superposition of microwaves.

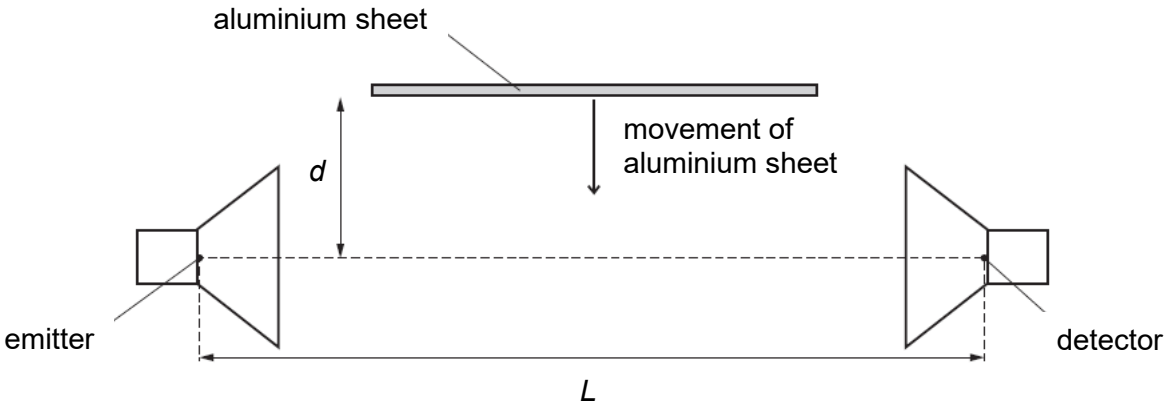


Fig. 4.3 (not to scale)

A microwave emitter and detector face each other with a distance *L* apart. A reflecting aluminium sheet is placed at a vertical distance *d* above the direct path between the emitter and the detector.

(i) When the aluminium sheet is moved in the direction shown by the arrow, the intensity of microwaves received by the detector fluctuates.

Describe and explain this variation in the microwave intensity.

Reducing d changes the path difference between the direct and reflected wave so the phase difference changes [B1] idea of path diff / phase diff between direct and reflected wave NOT stationary wave formed by the reflected wave travelling in opposite direction

Constructive interference leading to maximum intensity (occurs when path difference is a whole number of wavelengths) AND destructive interference leading to minimum intensity (occurs when path difference is odd number of half-wavelengths) [B1] idea of constructive and destructive interference leading to intensity variations stated [2]

(ii) As d decreases, the microwave detector has a maximum reading at d = 21.2 cm and the next maximum at d = 17.7 cm. The distance L stays constant at 80.0 cm.

Determine the wavelength λ of the microwaves.

$$\text{reflected path} = 2\sqrt{\left(\frac{L}{2}\right)^2 + d^2} \quad \text{[C1]}$$

Use of Pythagoras theorem to calculate reflected path

$$\text{path difference} = 2\sqrt{\left(\frac{L}{2}\right)^2 + d^2} - L$$

$$2\sqrt{\left(\frac{0.800}{2}\right)^2 + 0.212^2} - 0.800 = (n+1)\lambda \quad \text{---(1)}$$

$$0.1054 = (n+1)\lambda$$

$$2\sqrt{\left(\frac{0.800}{2}\right)^2 + 0.177^2} - 0.800 = n\lambda \quad \text{---(2)}$$

$$0.0748 = n\lambda$$

[M1] for correct path differences seen

$$\lambda = 0.1054 - 0.0748 = 0.0306 \text{ m} \quad \text{[A1]}$$

correct method leading to correct ans

λ = ... m [3]

(iii) Microwaves undergo a phase change of π rad when they reflect.

When d is much less than the wavelength of the microwaves, it is observed that the detector reading is a minimum.

Suggest why this observation demonstrates that microwaves must undergo a phase change on reflection from the aluminium sheet.

When d is much less, path difference becomes negligible or almost zero (as distance travelled by reflected wave is almost similar to incident wave) [B1] implication of small d on path difference seen

This is supposed to produce a maximum reading due to constructive interference, but since a minimum was detected, it implies a phase change of π rad had occurred due to reflection [B1] idea that supposed to be C.I. but D.I. instead so must be due to phase change of π rad [2]

[Total: 14]

3 A sound system in a large auditorium has two identical loudspeakers, S_1 and S_2, separated by 1.5 m. The amplitude of the sound emitted by each loudspeaker is proportional to the voltage across its terminals. A balance control is used to adjust the voltage supplied to each loudspeaker. The arrangement is shown in Fig. 3.1.

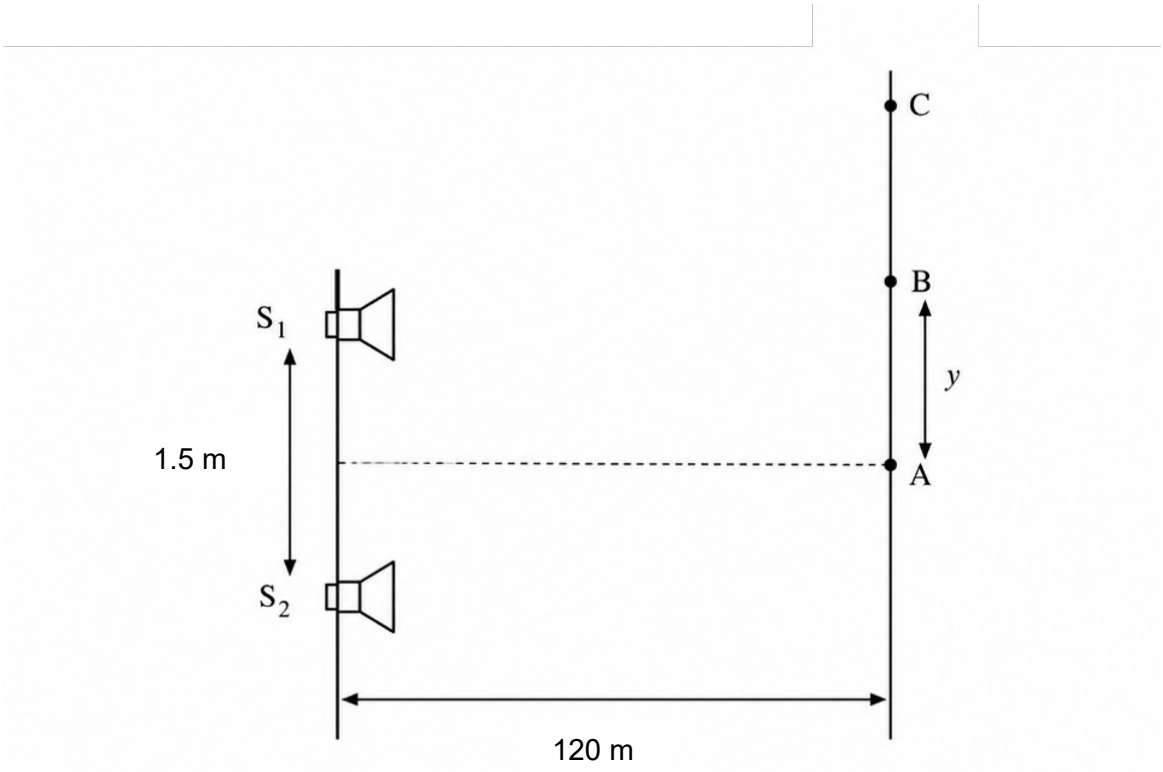


Fig. 3.1 (not to scale)

Initially, the loudspeakers emit sound waves of frequency 8500 Hz in phase. The balance control is adjusted so that the potential difference across each loudspeaker is 5.0 V r.m.s.

An observer at A hears a loud sound of intensity I_{max}. As he moves along the line AC, which is 120 m from the loudspeakers, the intensity first falls to zero at B, a distance y from A.

The speed of sound in air is 340 m s^{-1}.

(a) Determine the distance y.

$$\lambda = \frac{v}{f} = \frac{340}{8500} = 0.0400\,\text{m} \qquad [1]$$

Let the distance between the first two maxima be x.

$$x = \frac{\lambda D}{D} = \frac{0.0400 \times 120}{1.5} = 3.2\,\text{m} \quad [1]$$

Since B is the first minimum, the distance y is half of x.

$$y = \frac{x}{2} = \frac{3.2}{2} = 1.6\,\text{m} \quad [1]$$

y = ... m [3]

(b) Determine the next higher frequency of sound emitted by the speaker such that point B would also be a point of zero intensity.

The path difference at B is $\Delta x = \frac{\lambda}{2} = \frac{0.0400}{2} = 0.0200\,\text{m}$ [1]

For destructive interference, $\Delta x = (n + \frac{\lambda}{2})$

For the original frequency, this corresponds to n = 0. If the frequency is increased, it will correspond to n = 1 or $\Delta x = \frac{3\lambda'}{2}$

Hence

$0.0200 = \frac{3\lambda'}{2}$

$\lambda' = 0.01333\,\text{m}$ [1]

$f' = \frac{v}{\lambda'} = \frac{340}{0.01333} = 2.55 \times 10^4\,\text{Hz}$ [1]

frequency = ... Hz [3]

(c) The loudspeakers continue to emit sound of frequency 8500 Hz. The balance control is adjusted so that the voltages applied across S_1 and S_2 are 2.0 V r.m.s. and 8.0 V r.m.s. respectively. The amplitude of the sound emitted by each loudspeaker is proportional to the voltage applied across it.

In terms of I_{max}, determine the new intensity at B.

Initially, resultant amplitude at A, $A = A_1 + A_2 = 5 + 5 = 10$ units [1]

Later, the balance control is changed. At B, the waves meet out of phase:
$\therefore A' = A_1 - A_2 = 8 - 2 = 6$ units [1]

$\frac{I_{max}'}{I_{max}} = \frac{kA'^2}{kA^2} = (\frac{6}{10})^2$

$\therefore I_{max}' = 0.36 I_{max}$ [1]

Intensity = ...I_{max} [3]

Total: [9]